

Hwaran Lee

CONTACT INFORMATION	T-Brain at AI center, SK Telecom SK T-tower, Eulji-ro 65, Jung-gu, Seoul, South Korea, 04539	+82.10.2034.2420 — hwaran.lee@gmail.com hwaranlee.github.io
RESEARCH INTERESTS	Natural Language Processing/Understanding, Dialog Systems, Speech Recognition Systems, Generative models, Neural Networks and Machine Learning	
EDUCATION	KAIST Ph.D., Electrical Engineering Dissertation title: <i>Neural Representations for Speech Recognition and Natural Language Generation</i> Advisor: Prof. Soo-Young Lee Committee Members: Prof. Dae-Shik Kim, Hoirin Kim, Alice Oh, Jinwoo Shin, Sang-Hoon Oh	Daejeon, South Korea Mar. 2013 — Aug. 2018
	KAIST B.S., Mathematical Science Minor: Financial Engineering Program <i>Magna Cum Laude</i>	Daejeon, South Korea Feb. 2008 — Aug. 2012
RESEARCH AND WORK EXPERIENCES	Research Scientist - T-Brain, AI Center, SK Telecom, Seoul, South Korea Graduate Research Assistant - Electrical Engineering, KAIST, Daejeon, South Korea Undergraduate Researcher - Brain Science Research Center, Daejeon, South Korea	Nov. 2018 — Present Mar. 2013 — Aug. 2018 Jun. 2012 — Feb. 2013
PUBLICATIONS	International Journal [J1] Geonmin Kim, Hwaran Lee , Bo-Kyeong Kim, Sang-Hoon Oh, Soo-Young Lee, “Unpaired Speech Enhancement by Acoustic and Adversarial Supervision for Speech Recognition”, <i>IEEE Signal Processing Letters</i> , (2019): 159-163. [J2] Ho-Gyeong Kim, Hwaran Lee , Geonmin Kim, Sang-Hoon Oh, Soo-Young Lee, “Rescoring of N-best Hypotheses using Top-down Selective Attention for Automatic Speech Recognition”, <i>IEEE Signal Processing Letters</i> , (2018): 199-203. [J3] Hwaran Lee , Geonmin Kim, Ho-Gyeong Kim, Sang-Hoon Oh, Soo-Young Lee, “Deep CNNs Along the Time Axis With Intermap Pooling for Robustness to Spectral Variations”, <i>IEEE Signal Processing Letters</i> 23.10 (2016): 1310-1314. [J4] Hwaran Lee , Nadeem Iqbal, Wonil Chang, and Soo-Young Lee, “A Calibration Method for Eye-Gaze Estimation Systems Based on 3D Geometrical Optics”, <i>IEEE Sensors Journal</i> 13, no. 9 (2013): 3219-3225. [J5] Nadeem Iqbal, Hwaran Lee , Soo-Young Lee, “Smart User Interface for Mobile Consumer Devices Using Model-Based Eye-Gaze Estimation”, <i>IEEE Transactions on Consumer Electronics</i> 59, no. 1 (2013): 161-166. International Conference [C1] Gi-Cheon Kang, Junseok Park, Hwaran Lee , Byoung-Tak Zhang, and Jin-Hwa Kim, “DialGraph: Sparse Graph Learning Networks for Visual Dialog”, <i>Submitted</i> , 2020.	

- [C2] **Hwaran Lee***, Jinsik Lee*, and Tae-Yoon Kim, “SUMBT: Slot-Utterance Matching for Universal and Scalable Belief Tracker”, *The 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.
- [C3] Geonmin Kim, **Hwaran Lee**, Bo-Kyeong Kim, and Soo-Young Lee, “Compositional Sentence Representation from Character within Large Context Text”, *International Conference on Neural Information Processing (ICONIP)*, 2017.
- [C4] Ho-Gyeong Kim, Jihyeon Roh, **Hwaran Lee**, Geonmin Kim, and Soo-Young Lee, “Active Learning for Large-scale Object Classification: from Exploration to Exploitation” *In Proceedings of the 3rd International Conference on Human-Agent Interaction, (HAI)*, 2015.
- [C5] Bo-Kyeong Kim, **Hwaran Lee**, Jihyeon Roh, and Soo-Young Lee, “Hierarchical committee of deep CNNs with exponentially-weighted decision fusion for static facial expression recognition”, *In Proceedings of the 2015 ACM on International Conference on Multimodal Interaction (ICMI)*, 2015.

Conference without Proceedings

- [W1] Geonmin Kim*, **Hwaran Lee***, CheongAn Lee, Eunmi Hong, Byunggeun Kim, Soo-Young Lee, “A Deep Chatbot for QA and Chitchat.”, The Conversational Intelligence Challenge section on NIPS 2017 Competition Track Workshop, 2017.
- [W2] **Hwaran Lee**, Geonmin Kim, Jihyeon Roh, and Soo-Young Lee, “Learning Tonotopically Organized Auditory Feature-map from Speech by an Intermap Pooling Layer in a Deep CNN”, *15th China-Japan-Korea Joint Workshop on Neurobiology and Neuroinformatics (NBNI)*, 2015. (only abstract)
- [W3] Geonmin Kim, **Hwaran Lee**, Jaemyung Yu, and Soo-Young Lee, “Spoken Sentence Embedding from Character by Jointly Learning Character-level Compositional Word Model and RNN Sentence Encoder”, *15th China-Japan-Korea Joint Workshop on Neurobiology and Neuroinformatics (NBNI)*, 2015. (only abstract)

PATENTS

- [P1] Tae-Yoon Kim, Jin Kim, Hyungjoon Kim, Jinsik Lee, **Hwaran Lee**, Heewon Jeon, Seokhwan Jo, “Method and Apparatus for Providing Hybrid Intelligent Customer Consultation”, Korea Patent Application 10-2019-0136035, filed October 2019, Patent Pending.
- [P2] **Hwaran Lee**, Jinsik Lee, Tae-Yoon Kim, “Method and Apparatus for Dialogue State Tracking for Use in Goal-oriented Dialog System”, Korea Patent Application 10-2019-0086380, filed July 2019, Patent Pending.

RESEARCH PROJECTS

End-to-end goal-oriented dialog systems

2018 - 2019

- Funded by SK Telecom
 - Research and development of a scalable multi-domain natural language understanding
 - Research and development of end-to-end neural goal-oriented dialog systems
 - Participated in DSTC8 Track1 and ranked 6th place

A Deep Chatbot for QA and Chitchat

2017

- Funded by *Institute for Information & Communications Technology Promotion(IITP)*
 - Research and development of an article-based chatbot which can carry on both question-answering and chitchat conversations

*these authors contributed equally to this work

- Participated in *the Conversational Intelligence Challenge of NIPS 2017 Live Competition* **Ranked 3rd place**

Deep Learning based Korean Understanding and Applications 2015 – 2018

- Funded by HANCOM Inc.
 - Research and development of Korean morpheme- and syllable-level language models, sentence and document models
 - Research and development of continual learning for language models

Speech Feature Extraction based on Deep Learning for Continuous Speech Recognition Systems 2013 – 2014

- Funded by LG Electronics
 - Development of convolutional neural networks for acoustic models of continuous English speech, programmed C/C++ and CUDA based on KALDI toolkit

Development of Dialog-based Spontaneous Speech Interface Technology on Mobile Platforms 2013 – 2014

- Funded by Electronics and Telecommunications Research Institute (*ETRI*)
 - Development of deep belief networks and auto-encoders for acoustic models of Korean syllable data, and programmed models in C/C++ and CUDA languages

HONORS AND AWARDS	• Annual Roll Award, KAIST EE	Apr. 2018
	• Ranked 3rd, ConvAI challenge, NIPS 2017 Competition Track Workshop	
		2017
	• Challenge Winner, ICMI EmotiW2015	2015
	• Best Paper Award, HAI	2015
	• Qualcomm Innovation Award	2015
	• BK21 Plus Financial Support for Graduates Long Term Training	May. 2014
	• KAIST Graduate Scholarship	Mar. 2013 - Aug. 2018
	• Australian Endeavour Student Exchange Grant (AUD\$ 5000), The University of Queensland	Apr. 2011
	• National Excellence Scholarship, KOSAF	Feb.2008 - Feb. 2012
ACADEMIC PEER REVIEWS	• International Conference on Computational Linguistics (<i>COLING</i>)	2020
	• Samsung Humantech Paper Awards Committee	2020
	• Qualcomm Innovation Awards (Undergraduate Track) Committee	2019
	• Speech Communication	2019
	• IEEE Transactions on Neural Networks and Learning Systems	2017 - 2018
	• Neural Processing Letters	2015
INVITED TALKS	• Introduction to deep learning for dialogue systems Yeonsei Univ., Seoul, Jun. 2020	
	• Toward end-to-end neural dialog systems for multi-domain task completion KAIST, Daejeon, Dec. 2019	
	• Introduction to deep learning for dialogue systems Sookmyung Women's Univ., Seoul, Oct. 2019	
TEACHING EXPERIENCE	Teaching Assistant	
	• ChatBot PyTorch Implementation, KB-KAIST AI	Sep. 2017
	• EE476 Audio-Visual Perception Models	Spring 2016, 2017
	• EE538 Neural Networks	Fall 2015, 2016, 2017

Graduate Mentor

- Undergraduate Mentoring Program, KAIST Counselling Center
Fall 2014, Spring 2016

SKILLS

- **Languages** Python, Lua, MATLAB, C/C++, CUDA
- **Libraries** PyTorch, TensorFlow, Torch7, KALDI

REFERENCES

Prof. Soo-Young Lee (Ph.D. advisor)
Emeritus Professor, School of Electrical Engineering, KAIST sylee@kaist.ac.kr
Director, KAIST Institute for Artificial Intelligence <http://cnsl.kaist.ac.kr>